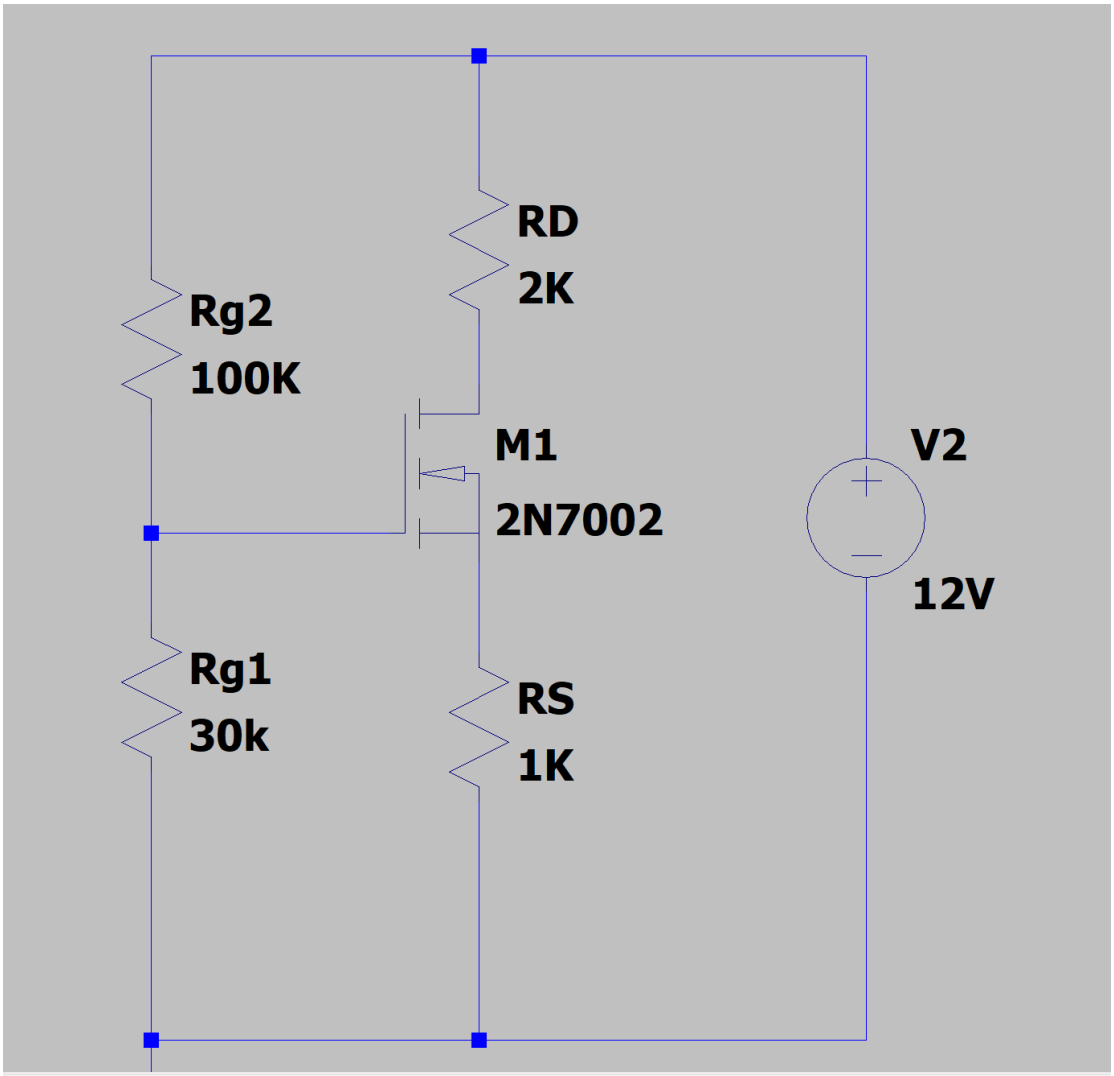
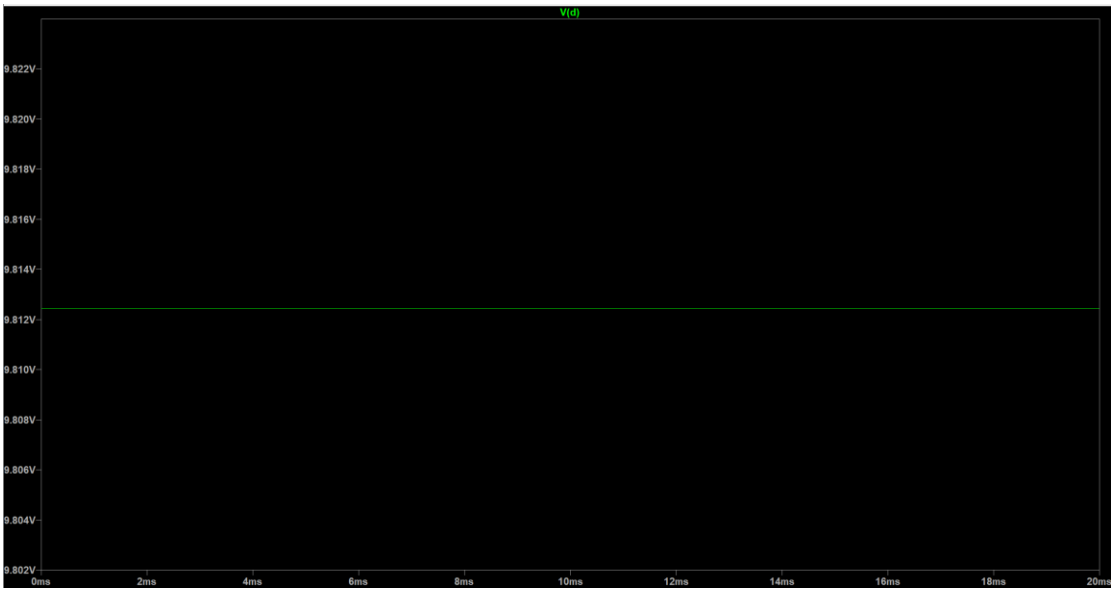
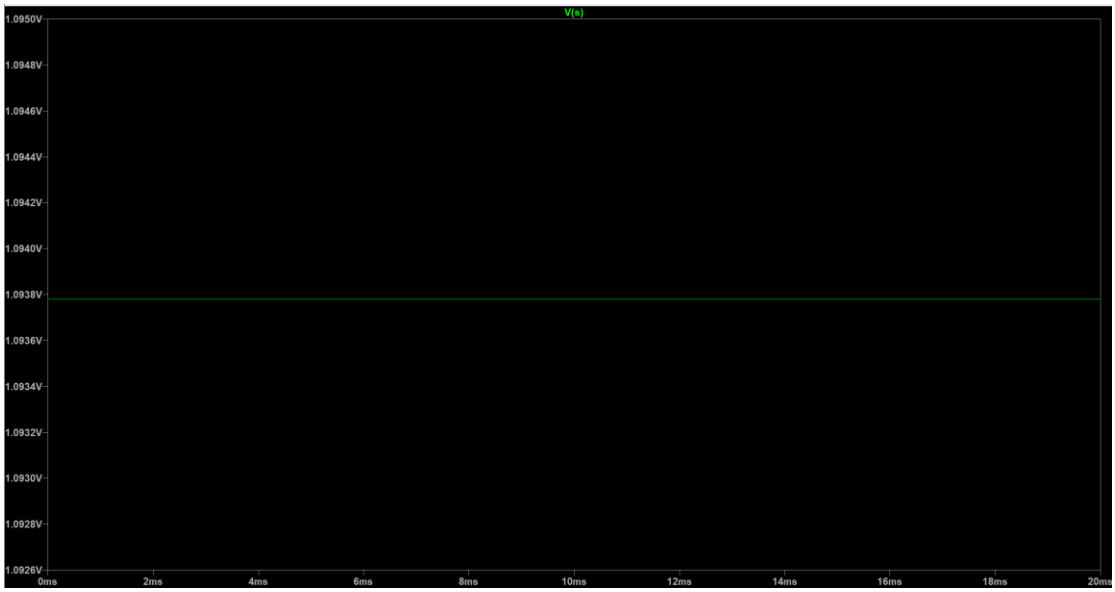
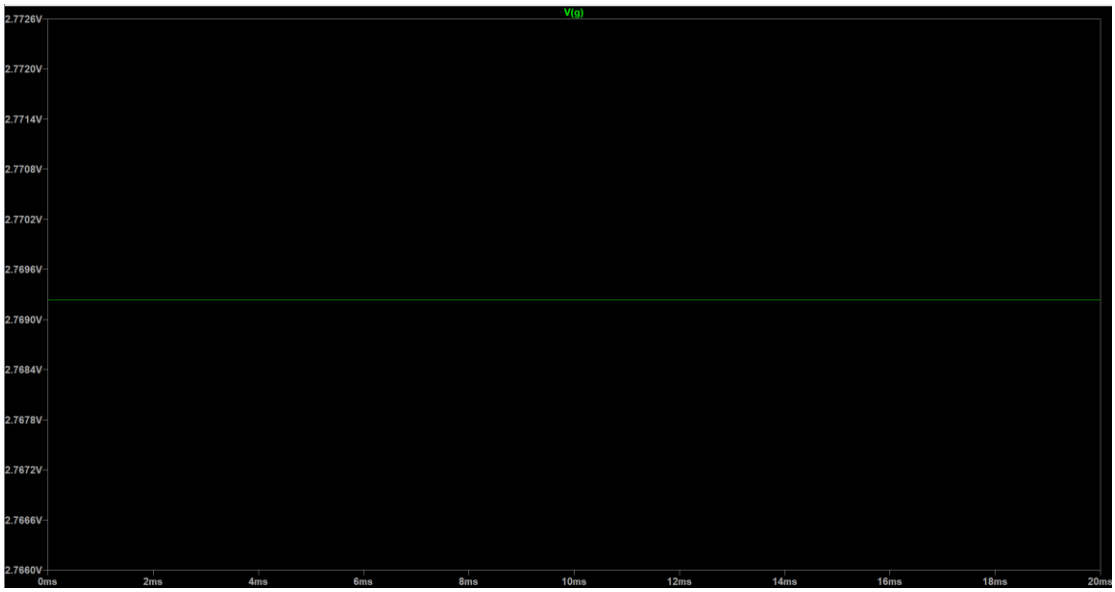
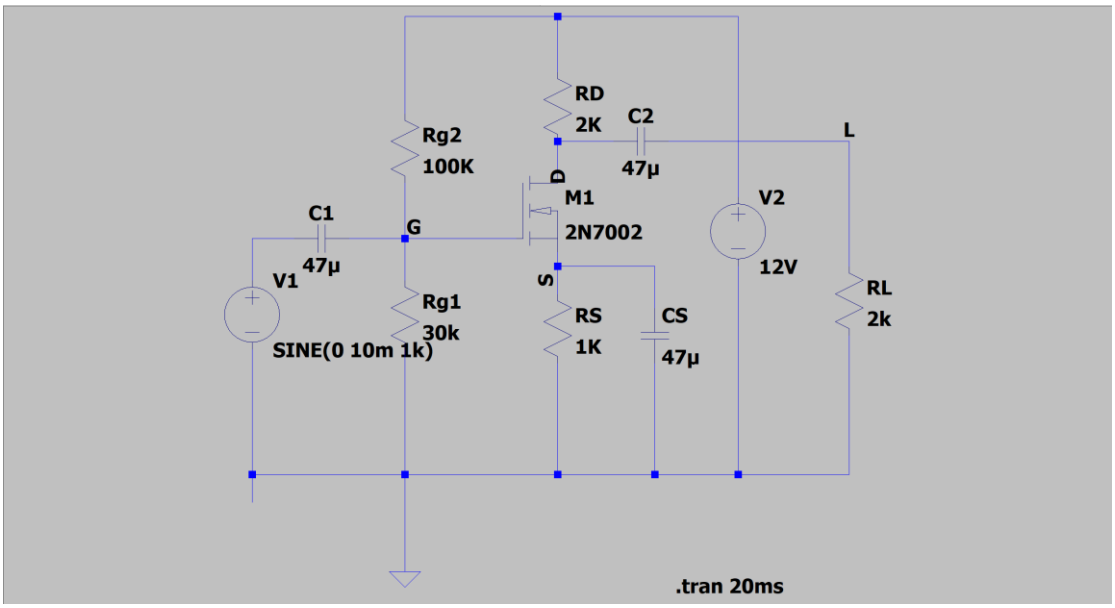

RD
2K
Rg2
100K
M1
2N7002
V2
12V
Rg1
30k
RS
1K

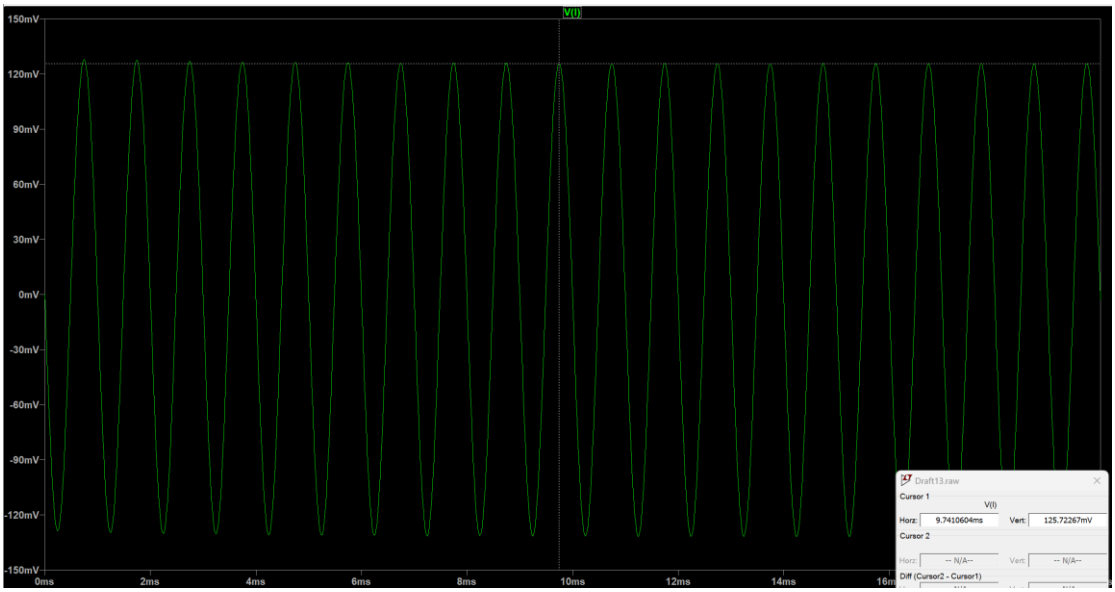

V(d)
9.822V
9.820V
9.818V
9.816V
9.814V
9.812V
9.810V
9.808V
9.806V
9.804V
9.802V
0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
16ms
18ms
20ms



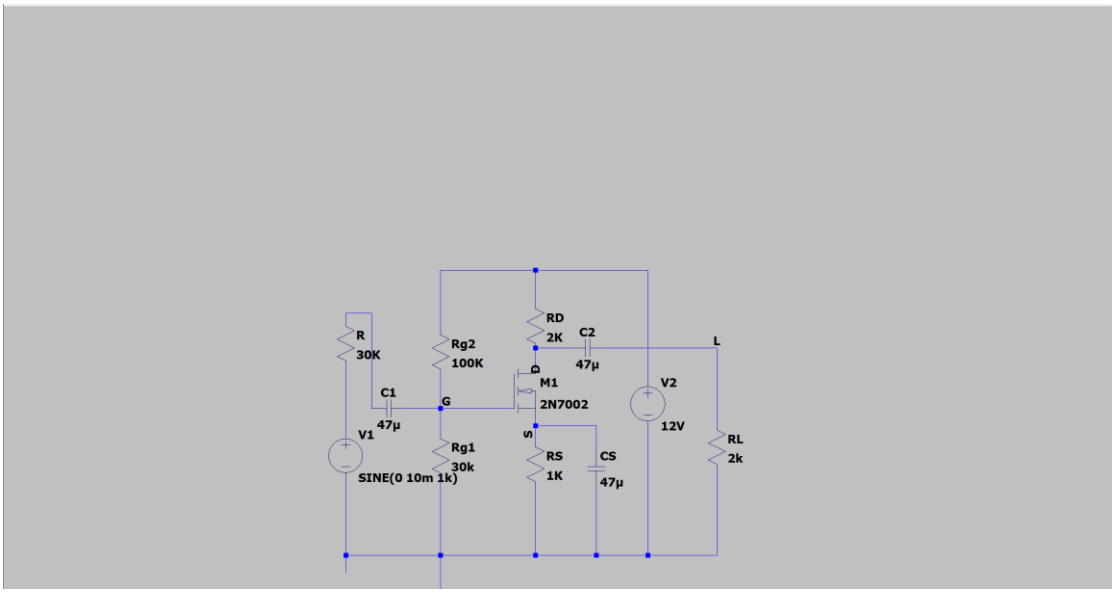


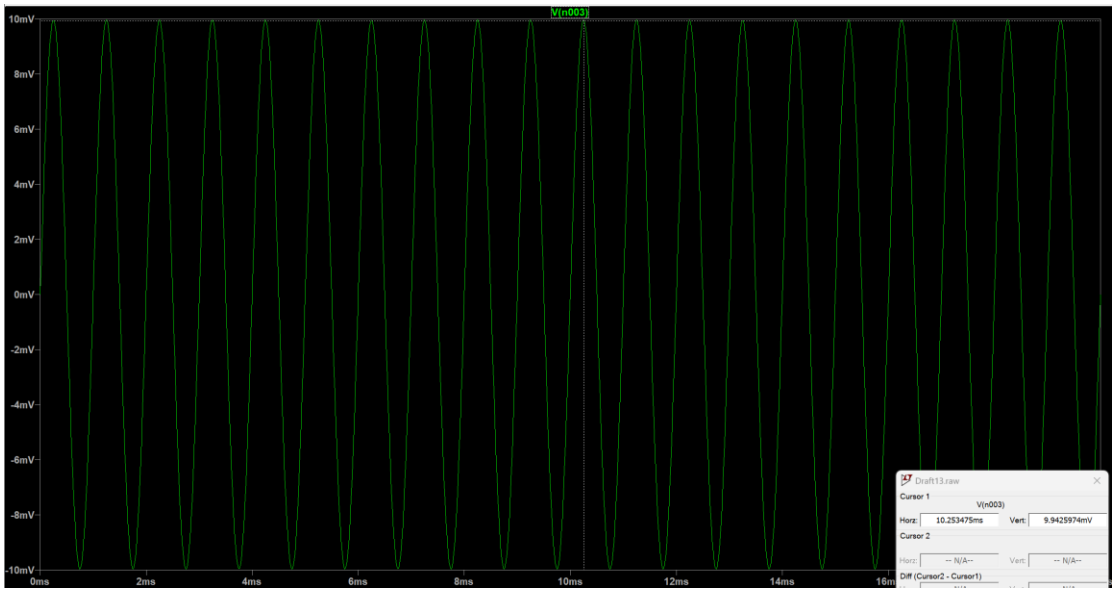
$$I_D = \frac{U_D}{R_D} = \frac{12V - 9.812V}{2000\Omega} = 1.094\text{mA}$$

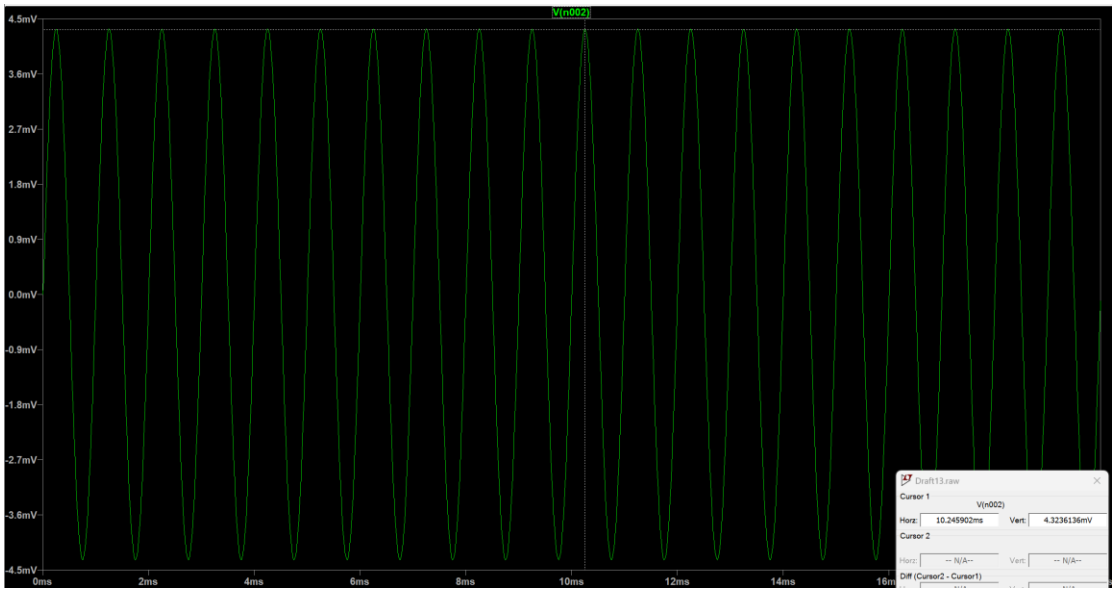

RD
2K
C2
47µ
L
Rg2
100K
D
M1
2N7002
V2
12V
C1
47µ
G
V1
S
Rg1
30k
RS
1K
CS
47µ
RL
2k
SINE(0 10m 1k)
.tran 20ms


V(l)
150mV
120mV
90mV
60mV
30mV
0mV
-30mV
-60mV
-90mV
-120mV
-150mV
0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
Draft13.raw
Cursor 1
V(l)
Horz: 9.7410604ms
Vert: 125.72267mV
Cursor 2
Horz: -- N/A--
Vert: -- N/A--
Diff (Cursor2 - Cursor1)

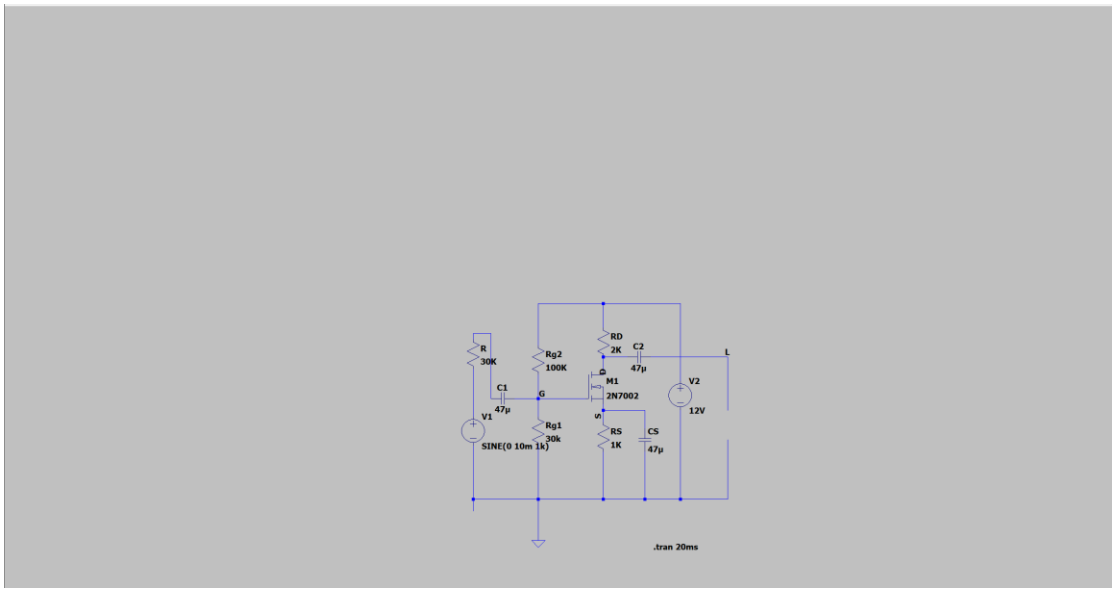
$$A_v = \frac{V_o}{V_i} = \frac{125mV}{10mV} = 12.5$$

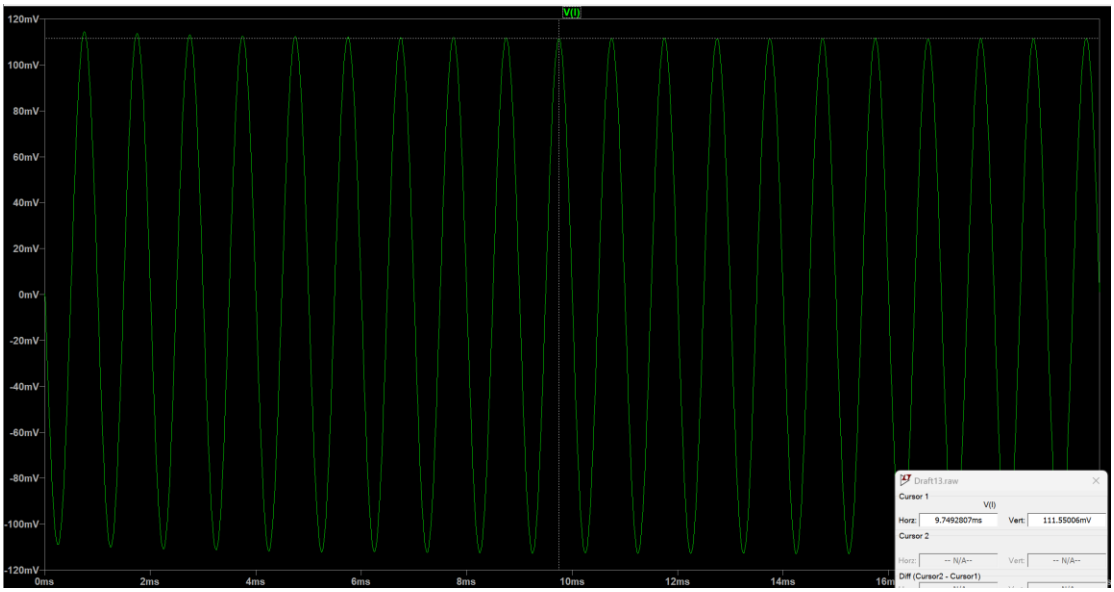


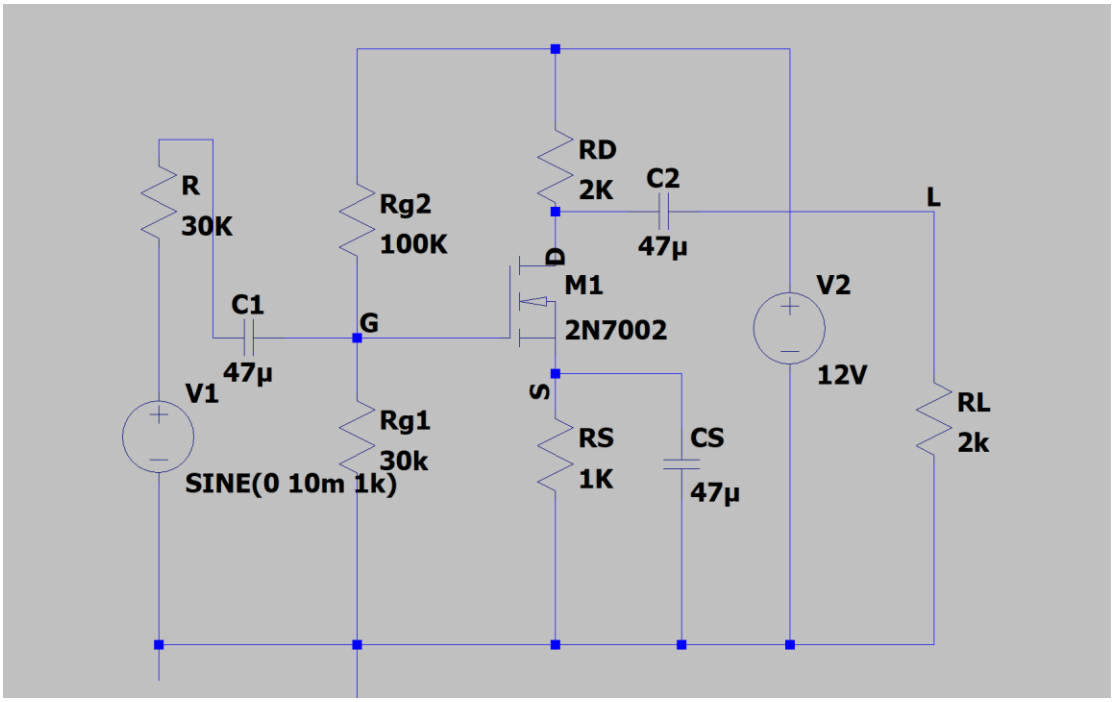


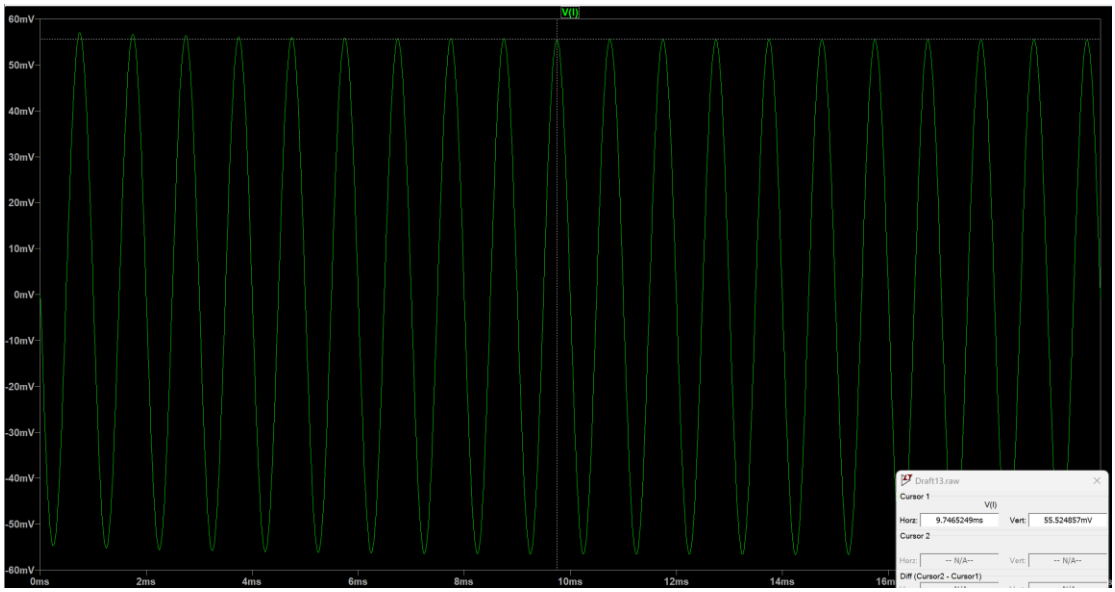


$$R_i = \frac{U_i}{U_s - U_i} \text{R} = \frac{9.94mV}{9.94mV - 4.32mV} \text{30K} = 53\text{K}\Omega$$




V(l)
120mV
100mV
80mV
60mV
40mV
20mV
0mV
-20mV
-40mV
-60mV
-80mV
-100mV
-120mV
0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
Draft13.raw
Cursor 1
V(l)
Horz: 9.7492807ms
Vert: 111.55006mV
Cursor 2
Horz: -- N/A--
Vert: -- N/A--
Diff (Cursor2 - Cursor1)


R
30K
Rg2
100K
RD
2K
C2
47µ
L
D
M1
2N7002
V2
12V
C1
47µ
G
V1
SINE(0 10m 1k)
Rg1
30k
S
RS
1K
CS
47µ
RL
2k

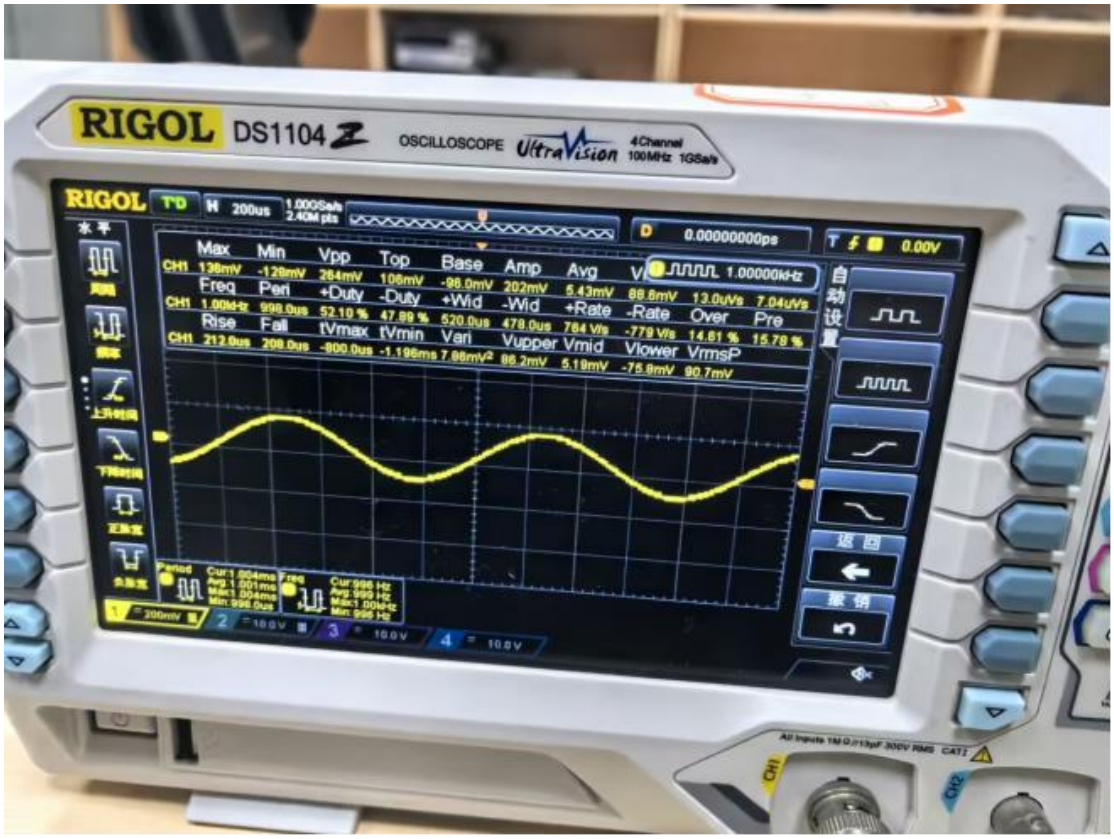


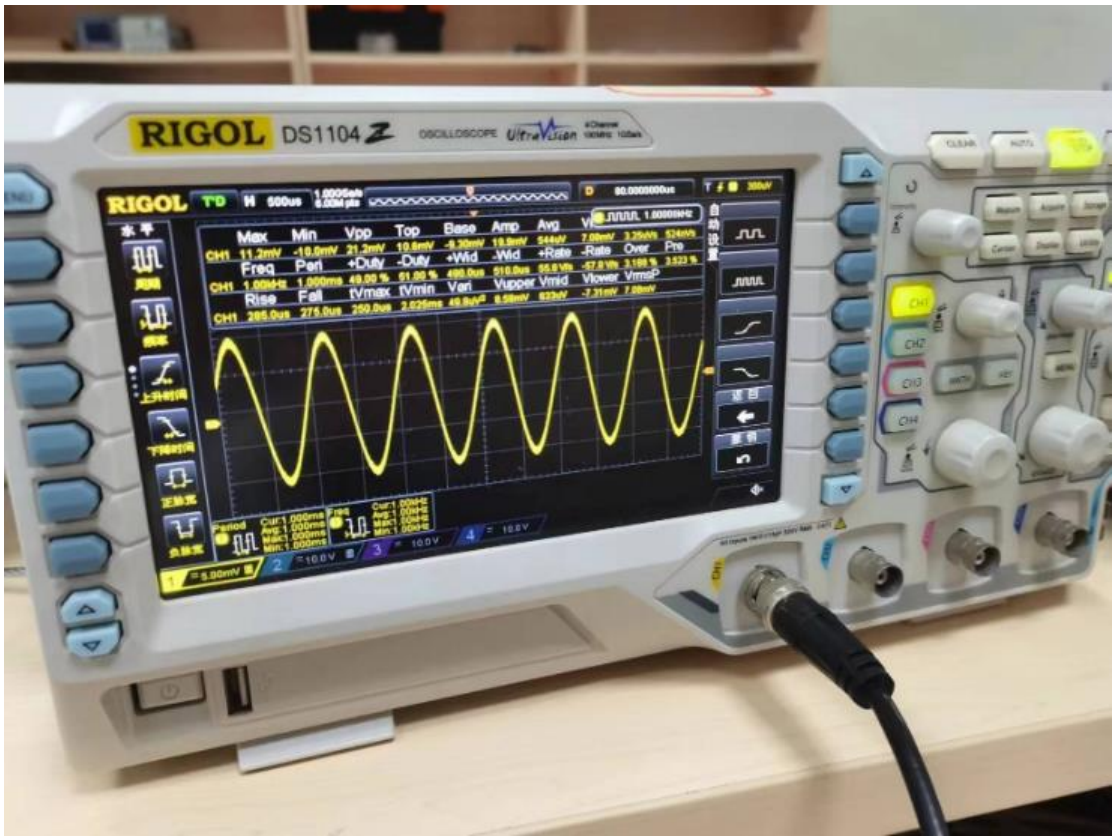
$$R_o = \left(\frac{U_o{}'}{U_o} - 1\right) R_L = \left(\frac{111mV}{55.5mV} - 1\right) R_L = 2\text{K}\Omega$$

quiescent point	voltage gain	input resistance	output resistance
U_{GSQ}=2.762mV I_{DQ}=1.094mA U_{DSQ}=9.812	$A_V = 12.5$	R_i=53KΩ	R_o=2KΩ

实　　物　　：

16V
Tantalum


RIGOL DS1104Z
OSCILLOSCOPE UltraVision 4Channel 100MHz 1GSa/s


RIGOL DS1104Z
OSCILLOSCOPE UltraVision